

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c = 3 \times 10^8 \text{ m/s}$		
$\lambda = \frac{c}{f} \text{ (m)}$	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
= 1	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$	
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$	$\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field	
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$	
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$.	
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D = 1.5$ (1.76 dB)	
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l	
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$	
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001 \text{ rad} \rightarrow \text{tiny} \rightarrow 0$.	
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$	

Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$	
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$	$S_0 = \frac{15I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$	
$D = 1.64$, $R_{\text{rad}} = 36.5 \Omega$, $\theta_{\text{max}} = \pi/2$.	
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, shortcut form	
$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$	
$D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\text{max}} = \pi/2$.	

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$$

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°	rad	sin θ	cos θ	sin ² θ	cos ² θ
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d	
elem pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$	
Equal phase ($\psi_i = 0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).	
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.	
Pattern multiplication $S_e = \text{single element pattern}$	
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$	
Array factor (general) $k = 2\pi/\lambda$	
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$	
Uniform $A_i = 1$, equal phase closed form	
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$	
Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.	
HPBW (broadside, uniform) use Nd , not $(N-1)d$	
$\beta \approx 0.88 \frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ \cdot \frac{\lambda}{Nd}$	
Grating lobes appear if $d > \lambda$ (broadside). Avoid.	

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377 \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
I_0 — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D$

FRIIS / dB

P_t, P_r — TX/RX power (W)
G_t, G_r — linear gains
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88 \lambda / \ell$ (rect/sq.)
$\beta \approx 1.02 \lambda / d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.88 \lambda / (Nd)$ (broadside)

BEAM PARAMETERS — β, Ω_p, D

Normalize always start here	
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)	
Beamwidth β at threshold X	
Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)	
<i>Gaussian-only shortcut ($F = e^{-a\theta^2}$). For non-Gaussian F, solve $F(\theta_X) = X$ numerically:</i>	
Threshold	$X = 10^{\text{dB}/10}$
HPBW (−3 dB)	0.5
−6 dB	0.25
−10 dB	0.1

Pattern solid angle Ω_p recognize pattern, look up

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi \text{ (sr)}$$

$$N\theta \text{ dep.: } \Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta.$$

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F = 1$)	$4\pi \approx 12.57$
Hertzian $\sin^2 \theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2 / \ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: [2nd][d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π .

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{\text{dB}} = 10 \log_{10} D$
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$.	
Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$	
$G = \xi D$	$G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.	

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$	$\lambda/4$ (gnd)	1.64	2.15	36.6
monopole				
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 / 8\pi \text{ W}$. $\lambda/4$ monopole (above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved. Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use $G_t, G_r = \text{these } D \text{ values (e.g. half-wave } G = 1.64)$.

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m ²)	
$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$	
Physical cross section wire dipole, diameter d	
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)	
Inverse: D from A_e	
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
Hertzian special case $D = 1.5$	
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2 \text{ (m}^2\text{)}$	
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.	

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.

3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t , SNR, etc.

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g. $D_{\text{dB}} = 10 \log_{10} D$; $G_{\text{dB}} = 10 \log_{10} G$;
 $SNR_{\text{dB}} = 10 \log_{10} (P_r/P_N)$.

dB linear	dB linear
0	1
3	2
6	4
7	5
−3	0.5
−10	0.1

Memorize: 3 dB $\leftrightarrow \times 2$, 10 dB $\leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver	
$P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23} \text{ J/K}$; T_{sys} system noise temp (K); B receiver bandwidth (Hz).	
Signal-to-noise ratio	
$SNR = \frac{P_{\text{rec}}}{P_N}$ (unitless)	$SNR_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $SNR > 10 \text{ dB}$ for usable signal.	

APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)			
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$			
Directivity from beamwidths any aperture			
$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$			
HPBW and D by aperture shape uniform illum.; β in rad			
Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$		$\beta_x \approx 0.88 \lambda / \ell_x$	$D \approx 4\pi / (\beta_x \beta_y)$
ℓ_y		$\beta_y \approx 0.88 \lambda / \ell_y$	$= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88 \lambda / \ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02 \lambda / d$	$D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$
Scaling rules $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$			
Change	New D	New β	
Double ($A_p \rightarrow 2A_p$)	area	$D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$	